Provable Sparse & Modular Attention

Design attention with provable efficiency and specialization guarantees — bridging mechanistic insight and formal theory.

Efficiency improvements to attention are usually empirical — they work in practice but lack guarantees. This phase requires you to prove upper and lower bounds on your sparse attention mechanism, then verify those proofs empirically on synthetic tasks where the true dependency structure is known. The rarest combination in attention research: working code + written proofs.

CORE TASKS

- — Create attention modules that attend only to relevant input partitions with provable sparsity guarantees.
- — Design synthetic datasets where the true dependency graph is known and verifiable.
- — Prove upper and lower bounds on FLOPs vs. approximation error for your sparse mechanism.
- — Implement adaptive partition selection that adjusts sparsity based on input complexity at inference time.

MATHEMATICAL FORMULATION

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Sparse attention: A_ij = 0 if j NOT in N(i) [N(i) = neighborhood of token i]
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Recovery guarantee: P[||A_hat - A||_F <= eps] >= 1-delta under s-sparse dependency
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FLOP bound: T(n, s) = O(n·s·log(n/delta)/eps^2) vs. O(n^2) dense
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Prove: EXISTS function f where sparse attn matches dense attn within eps if graph G is s-sparse
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INSTRUMENTATION

- — Compare modular vs. dense attention on known dependency graphs — compute recovery rate.
- — Track head specialization: mutual information between head outputs and known graph partitions.
- — Measure gradient flow efficiency through sparse paths vs. dense baselines.
- — Verify theoretical bounds empirically across sequence lengths and sparsity levels.

DELIVERABLES

Type	Description
Code	Sparse/modular attention layer with adaptive partition selection
Theory	Written proofs of FLOP bounds and approximation guarantees (LaTeX)
Benchmark	Synthetic task suite with known ground-truth dependency graphs
Writeup	Full paper-style report including failure mode analysis

RESEARCH INSIGHT

> *This bridges mechanistic understanding and formal guarantees — the rarest combination in attention research. Most efficiency work lacks proofs; most theoretical work lacks working implementations. You are building both.*

Difficulty: PhD-level | Sparse attention · Formal proofs · Complexity bounds · Graph structure